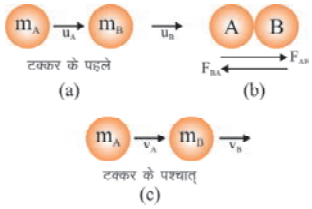
Chapter 5

Force and Laws of Motion



In a daily life we see different objects around us. We find that some objects appear to be stationary while others seem to be in motion. How do we bring any stationary object into motion? To ride a bicycle we need to pedal it. When we stop pedaling the speed of the bicycle decreases. To keep the cycle in motion we need to pedal again. Similarly, to bring a ball in motion, we can push or throw it. On applying a push, does the ball get in motion forever? or does it come to a stop in a while? What is the reason behind the ball to stop?

To bring a ball in motion, we need to push it and then it slowly comes to a halt. How does a ball in motion comes to rest? Does it stop on its own or is there a factor that causes it to stop? We have read earlier that to bring an object into motion or to bring it to rest we need an external factor which is called as 'force'.

Activity-1

In this activity we need a smooth ball made up of a wooden piece, two wooden planks of length 1 m. and 30 cm. each, a plastic tray, sand and some oil.

- By placing the smaller plank at 45° angle from the ground make an inclined plane (fig.1).
- Place the longer plank below the inclined plane, touching it as shown in the figure.
- Place the wooden ball at the top of the inclined plank and leave it.
- Measure the distance covered by the ball on the longer plank.
- Now replace the longer plank with a tray filled with sand.

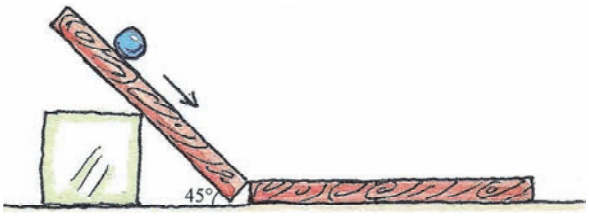


Fig. 1

- Again, place the ball at the top of the inclined plane and leave it. now measure the distance covered by the ball on the tray of sand.
- Similarly, repeat the above steps using a smooth plastic tray and some such objects which get smoother after applying oil.
- Measure the distance covered by the wooden ball on all of these surfaces.

Now try answering the following questions depending upon the above activity.

1. Did the ball cover same distance on all the surfaces.
2. Arrange the distance covered by the ball on different surfaces in a particular order.
3. In which situation did the ball cover maximum distance and why?

In the earlier classes, we have read about the frictional forces which are responsible for bringing the motion of an object to rest. The value of frictional force changes according to the nature of surfaces in contact. The ball covers larger distance on the surface on which it experiences lower frictional force.

On this basis, state which of the surfaces in the given activity applies the smallest frictional force?

As we saw, an object remains in a state of motion for much longer on the surfaces which apply lower frictional force.

What would happen if the frictional force on a surface becomes zero? Will the object ever stop?

According to Galeleo, in the absence of an external force of resistivity, an object moving at a constant velocity remains in its state of motion and similarly an object at rest remains in the state of rest until an external force works on it. Therefore, we say that an object remains in its state of motion or rest unless an external force brings a change in its state. This is known as 'inertia'. We can witness several examples of inertia in our daily lives. For example, on suddenly applying brakes of a moving vehicle, the person sitting inside it involuntarily bends in the forward direction and the items in the vehicle also slide forward. You must have felt this too. Give some more examples of such situations. Can you tell why does this happen? Discuss amongst yourself.

A person sitting inside a moving vehicle is in motion alongwith the vehicle. When the vehicle comes to a halt, the lower body of the person which is in contact with the seat comes to rest immediately while the upper body remains in motion due to inertia.

Hence, the person sitting inside the vehicle bends forward. Similarly you can tell why a person bends backwards when the vehicle starts to move suddenly. Discuss among yourself and write the answer in your notebook.

5.1 Inertia and Mass

From all the examples that are given uptil now, we can see that every object resists any change in its state, whether it is at rest or in motion. It does not want to change its state and wants to remain in its initial state of motion. This characteristic of an object is called 'inertia'. Several other questions arise from this. For instance we can ask whether all the objects have inertia? If yes then on what factors its value depends upon and is the inertia for all objects same? Come lets understand this through an activity.

Activity-2

- Take two plastic bottles 'A' and 'B'.
- Fill water in bottle 'B'.
- Drop both the bottles by pushing them with your hand.
- Which bottle fell easily.

You will find that it was easier to push and drop the empty bottle as compared to the filled one.

Similarly it is easier to push an empty carton box as compared to a carton filled with books.

We can think of several such other examples.

For example when there is just one person riding a bicyle, it is easier to bring it to a halt by applying brakes. But when a friend of your's sits behind you, it is difficult to apply brakes and bring the bicycle to a halt.

Can you explain the reason behind this? Discuss amongst yourself.

To change the state of a filled bottle, we need to apply more force as compared to an empty bottle. In a similar way, when a single person rider a bicycle, it is easier to apply brakes while in case of two people sitting on the bicycle, more force has to be applied. This means that heavier objects have more 'inertia'. Inertia is that property due to which an object tries to remain in its state of motion or state of rest. Mass is the measurement of resistance of an object towards any change in its state. Which means that mass is the measure of the property of inertia of any object. If the mass of an object is more, its inertia will be greater. The S.I. unit of mass is kilogram (kg.)

5.2 Balanced and Unbalanced Force

Activity-3

Take a block of wood. On two opposite faces of this block, put a hook or a nail on each. tie a thread A and B of similar lengths on both the hooks, as shown in the fig.2. Now place this block on the table Now, if we pull only the thread A by applying a force P then the block shifts towards the right side. If only the thread B is pulled using a force Q, the block shifts towards left. But if we apply similar forces on both the threads and pull them, then we see that the block does not move. These similar forces balance each other and so there is no change in the position of the block. The balanced force in this example is shown in the fig.3 (where P = Q).

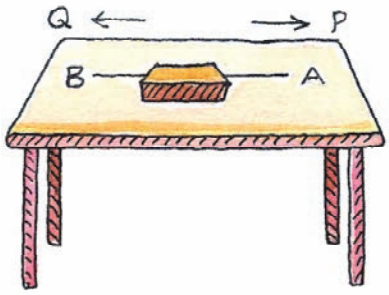


Fig. 2

If both the forces are not equal, then the net force on the block will be

= P – Q where P > Q

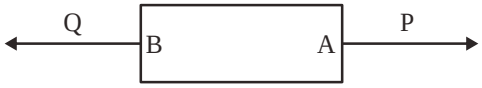


Fig. 3

Net force is the calculated force applied on the block. To understand net force, we will need to know all the forces that are applied on the object along with their directions. To get the net force we sum up all the forces that are applied in one direction and subtract all those forces that are applied in opposite direction. The net force applied on the block is a vector sum of P and Q. Therefore the net force applied on the block in a state of balanced force is-

Net force = ($\because$ P = Q)
= 0

If the force applied on the thread B is more than the force applied on A, then the block will shift towards the left side. Both the forces are not balanced and so the motion of the object will be towards the force with greater magnitude ie., towards B. (from figure number-3)

Force applied on the block = Q – P

5.3 First law of motion

Newton presented the laws of motion by understanding the relation between force and motion. According to Newton, "every object remains in its state of rest or a state of uniform motion until some unbalanced external force acts upon it."

So we have read, inertia is that property of an object due to which the object resists any change in its state. Therefore, the first law of motion is also known as the 'law of inertia'.

Can you explain the following examples according to the first law of motion?

- On beating a blanket with a rod, all dust particles get removed.
- On shaking the trunk of a tree, the fruits hanging on its branches drop down.

Explain the reason behind these after discussing with your friends. Can you think of some other such examples?

> When we jump on the same position in a moving train, we fall back on the same position again. Why does it happen so?

5.4 Linear Momentum

The first law of motion tells us that when an unbalanced external force acts on an object, then there is a change in the velocity of the object. This change in velocity depends upon the force applied on the object. But how can we get to know the change in velocity? Lets discuss on this further.

If a tenis ball and a cricket ball thrown at a high speed hits you, you will get more hurt by which of the two balls? If a cycle and a truck moving at the same velocity hit a wall, the wall will get destroyed more due to which of the two? Think of more such examples and discuss on it. To think more on this, do the following activity-4.

Activity-4

- Take a big tray.
- Fill it with sand or wet soil.
- Drop a cricket ball and a plastic ball simultaneously from a same height on the tray. What happens? Is the effect of both the balls similar on the sand.
- If the cricket ball is dropped on the tray from different heights, what will be its effect? When will the ball sink more into the soil?

This activity shows that the impact of an object depends upon its mass. The impact of a cricket ball having greater mass is more. We also see that the ball when dropped from a greater height, falls on the sand with higher speed and sinks into the sand more. Mainly, the impact depends upon the velocity of the object and its mass. If any of these two quantities is increased in similar canditions, then the impact also increases.

The product of the mass of an object and its velocity is called 'momentum'. The momentum 'P' of an object is defined by the product of its mass 'm' and velocity 'v'. That means, P = mv.

Momentum is a vector quantity which has both magnitude and direction. The direction of momentum will be similar to the direction of velocity. The S.I. unit of momemtum is kilogram metre/second (kg. m/s).

Force and the change in momentum

Think of such a situation where a car with a week battery is pushed on a road so as to move it in a straight line. The car does not start initially but on applying force constantly for some time, the speed of the car increases. Now the car's engine can be ignited. We can say that by constantly applying force, the momentum of the car also increases gradually. It is clear through this that the change in momentum of the car does not only depend upon the measure of the force but also depends upon the time for which the force is applied. The change in momentum of an object depends mainly upon the amount of force applied and the total time of application of force.

5.5 Second law of motion

According to the second law of motion, "The rate of change of momentum of an object is directly proportional to the unbalanced force applied on that object".

If F is the net unbalanced force applied on the object, P is the momentum, then according to the second law of motion,

$$F \alpha \frac{p_2 - p_1}{t_2 - t_1}$$

where P_1 is the initial momemtum at time t_1 and

p_2 is the final momemtum at time t_2.

$F \alpha \frac{\Delta p}{\Delta t}$ where $\Delta p = p_2 - p_1$ (change in momentum)

$\Delta t = t_2 - t_1$ (change in time)

To remove the sign of proportionality, we put here a constant of proportionality K.

$$F = K \frac{\Delta p}{\Delta t}$$

On taking K = 1, $F = \frac{\Delta p}{\Delta t}$

$$F = \frac{\Delta p}{\Delta t}$$

$$F = \frac{m\Delta v}{\Delta t} \quad [p = mv]$$

$$F = m\frac{\Delta v}{\Delta t} \quad \left[\because a = \frac{\Delta v}{\Delta t}\right]$$

$$F = ma$$

This is the mathematical representation for the second law of motion.

If m = 1 kilogram (1 kg.)

a = 1 meter/second^2 (1 m/s^2)

then, F = 1 kilogram × 1 meter/sec^2 (kg.m/s^2)

f = 1 Newton (N)

1 Newton force is that force which when applied on an object of 1 kg. produces an acceleration of 1 m/sec^2 to the motion. Newton is the S.I. unit of force, it is represented using N.

Example-1. The force applied on an object of 3 kg. changes its velocity from 2 m/sec. to 3.5 m/sec. in a time period of 25 sec. Find the measure of the force applied.

Solution :

Mass of object	m	=	3 kg.
initial velocity of object	u	=	2 m/s
final velocity of object	v	=	3.5 m/s
time of application of force	t	=	25 sec.
to find,	F	=	?

we know that $F = \frac{p_2 - p_1}{t} = \frac{mv - mu}{t}$

$$F = \frac{m(v-u)}{t}$$

$$F = 3\frac{(3.5-2)}{25}$$

$$F = \frac{3\times1.5}{25} = \frac{3\times15}{250} = \frac{45}{250}$$

$$F = 0.18\text{N}$$

Therefore, the measure of applied force will be 0.18 Newton.

Example-2. On applying some force on a trolley of 5 kg. an acceleration of 10 m/s^2 is produced. Find the applied force.

Solution :

Mass of the trolley	m	=	5 kg
acceleration produced on the trolley	a	=	10 ms^{-2}
Force applied on the trolley	F	=	?
	F	=	ma
	F	=	5 × 10
	F	=	50 N

Force applied on the trolley is 50 N.

Example-3. The figure shows the velocity-time graph of a 30 gm ball moving in a straight line on a table. How much frictional force needs to be applied to bring the ball to a state of rest.

Solution :

initial velocity of the ball u = 15 cm/s = 0.15 ms^{-1}

final velocity of the ball v = 0

time t = 5 s

mass of the ball m = 30 gm = 0.03 kg

Frictional force applied on the ball F = ?

$$F = m\frac{(v-u)}{t}$$

$$F = \frac{0.03\times(-0.15)}{5} = -0.0009 \text{ N}$$

$$F = -9\times10^{-4} \text{ N}$$

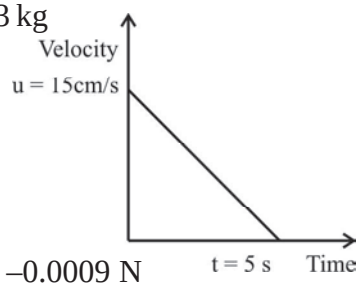


Fig. 4

Here, the negative sign shows that the frictional force is applied in an opposite direction to the direction of the initial velocity.

Example-4. A force of 10 Newton is applied for 5 sec. on an object of 5 kg. How much acceleration is produced due to this force? Also calculate the change in velocity of the object.

Solution : We know that the mass of the object $m = 5$ kg.

Force applied on the object $F = 10$ N

Time $t = 5$ s

We want to know the acceleration a and and the change in velocity $v - u = ?$

as, $F = ma$

that means, $10 = 5 \times a$

$a = 2 m/s^2$

$\because a = \frac{v-u}{t}$

$2 = \frac{v-u}{5}$

$v - u = 2 \times 5$ m/s

$v - u = 10$ m/s

Change in velocity , $v - u = 10$ m/s

Discuss

1. Why does a cricketer pull his hands backwards while catching a ball coming at a high speed?
2. When we jump from a height on a cemented floor we get hurt more as compared to when we jump from the same height on a bundle of hay.
3. A body moving at a constant speed is made to stop in 0.25 sec. by applying a force of 200 N. Calculate the initial momentum of the body.
4. A force of 5 N leads to an acceleration of 8 m/s^2 on a body of mass m_1. Same force causes an acceleration of 24 m/s^2 on another body of mass m_2. If both the bodies are tied together, how much acceleration will be produced by the same force?
5. If the net unbalanced force becomes zero, will the momentum also become zero? Explain.

5.6 Third law of motion

To understand the third law of motion, let us understand the following activity.

Activity-5

- Take a spring balance A and hang its fixed end on a wall with the help of a nail.
- Pull the free end of the spring balance and note the force applied.

- Now connect another spring balance B with A, as shown in fig.5.
- Pull both the springs A and B together and note down the applied force.
- You find that both the spring balances show the same value of force. Why is it so?

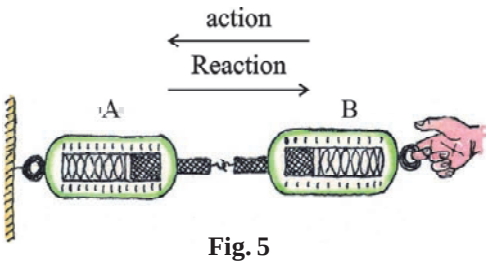


Fig. 5

In the above example, the spring balance B exerts a force on spring balance A. And a similar force is applied by A on B but in the opposite direction. Thus, both the similar forces act upon two different objects. When one body applies a force on the second body, the second body also applies the same amount of force on the first body but in the opposite direction. These pairs of forces acting on bodies are called as action-reaction forces.

According to the third law of motion "every action has an equal and opposite reaction". Action and reaction forces always act on two different bodies. Therefore even when these forces are equal and opposite, the net force is not equal to zero.

What happens when you walk? To start walking you need some force which causes acceleration. To produce acceleration we apply force on the ground through our foot, in a direction opposite to the direction of our movement. The ground also applies the same amount of force on our foot in an opposite direction which makes us move forward.

Notice that even if both the action and reaction forces have same magnitude, they do not cause same acceleration as the masses of both the bodies on which action-reaction forces act are different. The earth does not start moving backwards on getting pushed by our feet though it produces acceleration required by us to move forward.

If a body A applies a force of F_1 on body B and body B applies a force F_2 on body A, then according to the third law of motion $F_1 = -F_2$.

The negative sign here shows that the direction of force F_2 is opposite to the direction of F_1.

The first and second laws of motion are applied on the same object while the third law of motion is applied on two objects working together. Action and reaction represent those forces which act on two different objects at the same time.

Think about some other example of third law of motion.

5.7 The law of conservation of momentum

Suppose you have two balls A and B which have masses M_A and M_B respectvely. These two move in a straight line at a velocity of u_A and u_B respectively. There is no unbalanced force acting on them. Both

these balls hit each other after a time t, the force applied by A on B is F_{AB} and the force applied by B on A is F_{BA}. Thus, the velocity of A and B now becomes V_A and V_B respectively.

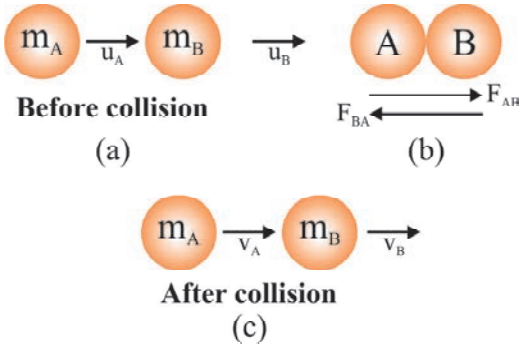


Fig. 6

The force applied by each ball on each other is equal and opposite, according to third law of motion.

Hence $F_{AB} = -F_{BA}$

There is a change in velocities of the balls due to these applied forces and hence change in momentum takes place.

So, according to second law of motion

Force = rate of change in momentum.

$$F_{AB} = \frac{m_B(v_B - u_B)}{t}$$

$$F_{BA} = \frac{m_A(v_A - u_A)}{t}$$

From third law of motion,

$$F_{AB} = -F_{BA}$$

$$\frac{m_B(v_B - u_B)}{t} = \frac{-m_A(v_A - u_A)}{t}$$

$$m_B(v_B - u_B) = -m_A(v_A - u_A)$$

$$m_Bv_B - m_Bu_B = -m_Av_A + m_Au_A$$

$$m_Av_A + m_Bv_B = m_Au_A + m_Bu_B$$

Momentum after collision = momemtum before collision

Through this equation we find that when there is an external force applied on the balls then there total momentum before and after collision remains same. The total momentum of both the balls does not change though their individual momentum does change. This is known as the law of conservation of momentum.

According to this law "The total momentum of two objects remains conserved in the absence of an external unbalanced force".

Let us perform an activity for this.

Activity-6

- Take a huge balloon and after filling air in it tie its mouth using a thread.
- Attach a straw on the surface of the balloon using a cellotape.
- Pull out a thin and long thread accross the straw.
- Now tie both the ends of this thread across the walls. (as in fig.7)
- Now untie the thread put on the mouth of the balloon.
- As soon as you untie the thread, the air filled inside the balloon gushes out.
- Observe the motion of the balloon and the straw.

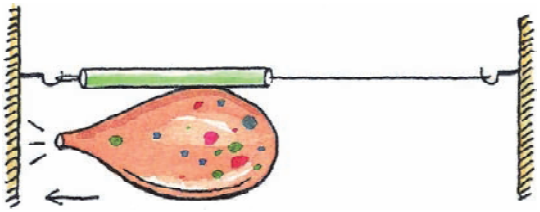


Fig. 7

You can see that the third law of motion is applied in this situation. But can you say which are those objects on which this law is being applied?

Find out more examples which work on this similar concept.

Example-5. Two bodies of 40 kg. and 20 kg. masses are moving at a velocity of 10 m/sec and 50 m/sec. Their direction are towards each other. After a certain time period, they collide and stick to each other. What will be the velocity of this combined object.?

Solution : Given that,

Mass of first spherical object is $m_1 = 40$ kg

Mass of second spherical object is $m_2 = 20$ kg

Velocity of first spherical object $u_1 = 10$ m/s

Velocity of second spherical object $u_2 = 50$ m/s

Velocity of combined object after collision $v = ?$

According to law of conservation of momentum

$$(m_1 + m_2)\, v = m_1u_1 + m_2u_2$$

$$(40 + 20)\, v = 40 \times 10 + 20 \times (-50)$$

$$60v = 400 - 1000$$

[$\because$ As objects are moving towards each other $u_2 = -50$ m/s]

$$v = -\frac{600}{60}$$

$$v = -10 \text{ m/s}$$

Velocity of the combined object after collision is 10m/s and it will move towards the direction of 20 kg. object.

Example-6. A bullet of 20 gm is shot from a gun of 2 kg. at a velocity of 100 m/s. Find out the recoil velocity of the gun.

Solution :

mass of the gun $m_1 = 2$ kg

initial velocity of the gun $u_1 = 0$

mass of the bullet $m_2 = 20$ gm = 0.02 kg

initial velocity of the bullet $u_2 = 0$

final velocity of gun $v_1 = ?$

final velocity of bullet $v_2 = 100\ m/s^{-1}$

According to law of conservation of momentum

$$m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$$

$$2 \times v_1 + 0.02 \times 100 = 20 \times 0 + 0.020 \times 0$$

$$2v_1 + 2 = 0 \quad \text{or} \quad 2v_1 = -2$$

$$v_1 = -1\ m/s$$

Therefore, the gun will recoil with a velocity of 1 m/s backwards.

Discuss

1. If the action and reaction are always equal then explain how a bull is able to pull a cart forward?
2. The motion of a rocket is based on which concept?
3. A fire extinguisher is hurt by a rubber pipe that throws water at a very high speed. Explain this.

Questions

1. Two bodies of masses 1 kg. and 2 kg. are moving parallel to a line in the same direction, at a velocity of 2 m/s and 1 m/s respectively. Both the bodies collide with each other. After collision, the velocity of the first body becomes 1.5 m/s. Find the velocity of the second body after collision.



What we have learnt

- Force is that external factor which causes a change in the state of motion or the state of rest of an object or tries to cause a change.
- According to the first law of motion "A body remains in its state of rest or state of motion unless an external unbalanced force acts upon it".

- That property of an object due to which it resists any change in its state of rest or state of motion is called as inertia. The measure of inertia of an object depends upon its mass.
- The product of the mass and velocity of an object is called momentum. The direction of momentum is same as that of the direction of velocity of that object. The unit of momentum is kg ms^{-1}.
- If the time required for the change in momentum is less, then the applied force will be greater. In such a situation bringing a moving object to rest can hurt someone.
- According to the second law of motion, the rate of change of momentum of an object is proportional to the unbalanced force applied on it and is in the same direction.
- S.I. unit of force is $kgms^{-2}$ or Newton.
- According to the third law of motion, every action has an equal and opposite reaction. Both the action and reaction act on two different bodies.
- If there is no external force acting on an object then the momentum of the object remains conserved.

Keywords :- Balanced force, Unbalanced force, Inertia, Linear momentum, Laws of conservation of linear motion.

Exercise

1. Choose the correct option-

(i) The reason for inertia of a body is-

(a) only mass (b) only velocity

(c) mass and velocity both (d) None

(ii) A boy is sitting on the uppermust berth of a train. Just when the train is coming to a halt, he drops a mango in the open hands of his brother who is sitting on the seat right below him. Where will the mango fall?

(a) right in his brother's hand

(b) Some distance away from his hand in a direction opposite to the direction of train.

(c) Some distance away from his hands in the direction of the train.

(d) None of these.

(iii) The concept of rocket launching is based on-

(a) First law of motion (b) Second law of motion

(c) Law of conservation of mass (d) Law of conservation of momentum

(iv) A ball stops in 0.06 sec. when a force of 500 N is applied to it. The momentum of the ball is-

(a) 500 N (b) 500 kgm/s (c) 30 kgm/s (d) 30 N

2. Fill in the blanks-

(i) The law of conservation of momentum is applied in the absence of .....

(ii) Both the ends of a balance are pulled with a weight of 20-20 kg. on each side. The reading on the balance is .....

(iii) The first law of motion is also known as the law of .....

(iv) If the acceleration of a body is doubled while keeping its mass constant, the force becomes .....

(v) Force is a ..... quantity.

Discuss and Write

1. Three solids which are made up of aluminum, steel and wood respectively have same shape and volume. Say which of these solids will have more inertia and why?
2. Utensils made of glass break on falling on the ground but not when they fall on sand. Why is it so?
3. Write down the mathematical representation of the second law of motion and write the meaning of all the symbols used in it.
4. Why does a person get a backward push on using a gun.
5. Why is the luggage kept on the roof of a bus tied up with a rope?
6. Prove that if the momentum of two bodies is same then the velocity of lighter body will be more than the velocity of heavier body.
7. According to the third law of motion, when we apply a force on a body, the body applies equal and opposite force on us. If that body is a car standing aside a road, probably it will not get in motion on the application of a force by us. A student argues that both the forces here are equal and opposite and hence they cancel each other. Express your thoughts regarding this argument and tell why the car does not move in the opposite direction.
8. The velocity of a body of mass 50 kg. moving at a uniform accelerration becomes 7 m/s from 4 m/s in 6 sec. Calculate the initial and final momentum of the body and the force applied on it.
9. A football and a tenis ball are in a similar motion with same velocity. Both of them collide head on with each other and after some time they come to a stop. If the time interval of collision is 1 sec, then

(i) Which of the two balls will be more affected by the force?

(ii) The momentum of which ball will change more?

(iii) The acceleration of which ball will be greatest?

10. Prove that the total momentum of two objects remains constant before and after their collision.

11. The force-time graph of a body of mass 3 kg. is drawn in the following figure. Find the momentum of the body during the time interval of 0 to 10 sec.

